

differentiable in θ at the data-governing parameter $\theta \in \Theta \subset \mathbb{R}^p$ and Θ open with non-empty interior. Furthermore, assume that the (peri-null) prior densities $\pi(\theta | \mathcal{H}_0)$ and $\pi(\theta | \mathcal{H}_1)$ assign positive mass to a neighborhood at the data-governing parameter θ and are four times continuously differentiable at θ ; then $\text{BF}_{10}(Y^n) \xrightarrow{\mathbb{P}_q} \frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)}$. $\diamond$

Proof The condition that the model is Laplace-regular allows us to employ the Laplace method to approximate the numerator and the denominator of the peri-null Bayes factor by

$$p(Y^n | \mathcal{H}_j) = f(Y^n | \hat{\theta}) \left(\frac{2\pi}{n} \right)^{\frac{p}{2}} |\hat{I}(\hat{\theta})|^{-\frac{1}{2}} \pi(\hat{\theta} | \mathcal{H}_j) \times \left(1 + \frac{C^1(\hat{\theta} | \mathcal{H}_j)}{n} + \frac{C^2(\hat{\theta} | \mathcal{H}_j)}{n^2} + \mathcal{O}(n^{-3}) \right), \quad (7)$$

where $C^1(\hat{\theta} | \mathcal{H}_j)$ and $C^2(\hat{\theta} | \mathcal{H}_j)$ for $j = 0, 1$ are bounded error terms of the Laplace approximation (cf. Kass et al., 1990) and given explicitly by Eq. (32) and Eq. (33) based on the notation introduced in Appendix A.

From the fact that the peri-null and the alternative models define the same likelihood function, thus, have the same maximum likelihood estimator, and only differ in how the priors concentrate on the parameters, we conclude that

$$\text{BF}_{10}(Y^n) = \frac{\pi(\hat{\theta} | \mathcal{H}_1) \left[1 + \frac{C^1(\hat{\theta} | \mathcal{H}_1)}{n} + \mathcal{O}(n^{-2}) \right]}{\pi(\hat{\theta} | \mathcal{H}_0) \left[1 + \frac{C^1(\hat{\theta} | \mathcal{H}_0)}{n} + \mathcal{O}(n^{-2}) \right]}. \quad (8)$$

Identifiability and the regularity conditions on the model imply that the maximum likelihood estimator is consistent, thus, $\hat{\theta} \xrightarrow{\mathbb{P}_q} \theta$ (e.g., van der Vaart, 1998, Chapter 5). As all functions of $\hat{\theta}$ in Eq. (8) are smooth at θ , the continuous mapping theorem applies and the assertion follows. $\square$

Theorem 1 implies that $\text{BF}_{10}(Y^n)$ is inconsistent; for all data-governing parameter values with a neighborhood that receives positive mass from both priors, the peri-null Bayes factor approaches a limit that is given by the ratio of prior densities evaluated at the data-governing θ as n increases. Note that this holds in particular for the test point of interest, e.g., $\delta = 0$, which has a neighborhood that the peri-null prior assigns positive mass to. This inconsistency result can be intuited as follows. The peri-null Bayes factor compares two marginal likelihoods with the same data-distribution (or sampling) model, but different prior distributions on the same parameter space; hence, the peri-null Bayes factor effectively assesses which prior performs best, and this should not matter asymptotically (i.e., as the data accumulate, the posterior distributions of the two models converge, and consequently the change in the Bayes factor will converge as well).³

From Theorem 1 it follows that the Bayes factor comparing the alternative $\mathcal{H}_1 : \delta \in \Delta = \mathbb{R}$ against a directed hypothesis, say, $\mathcal{H}_+ : \delta > 0$, is also inconsistent. For data under any $\delta > 0$, the associated Bayes factor $\text{BF}_{1+}(Y^n)$ then

³ We thank the first anonymous reviewer for providing this intuition.

converges in probability to $\Pi_u(\{\delta > 0\})/\Pi_u(\Delta)$, where $\Pi_u(B) = \int_B \pi_u(\theta) d\theta$ with π_u the unnormalized prior on δ .

The limit described in Theorem 1 can also be derived differently. For instance, Theorem 1 (ii) of Dawid (2011) can be applied twice: once to approximate the logarithm of the marginal likelihood of the alternative model, and once for the null model.⁴ Another way to derive the limit in Theorem 1 is by using the generalized Savage-Dickey density ratio (Verdinelli and Wasserman, 1995) and by exploiting the transitivity of the Bayes factor. Theorem 1, however, can be more straightforwardly extended to characterize the asymptotic behavior of the peri-null Bayes factor.

The limiting value of the peri-null Bayes factor is not representative when n is small or moderate. Theorem 2 below shows that the sampling mean of $\log \text{BF}_{10}(Y^n)$ is expected to be of smaller magnitude than its limiting value. In other words, the limit in Theorem 1 should be viewed as an upper bound under the alternative and a lower bound under the null.

This theorem exploits the fact that without a point-null hypothesis the gradients of the densities $\pi(\theta | \mathcal{H}_1)$ and $\pi(\theta | \mathcal{H}_0)$ are of the same dimension, which implies that the gradient $\frac{\partial}{\partial \theta} \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right)$ is well-defined. As such, the delta method can be used to show that the peri-null Bayes factor inherits the asymptotic normality property of the MLE.

To state the theorem we write D for the differential operator with respect to θ , e.g., $[D^1 \pi(\theta | \mathcal{H}_j)] = \frac{\partial}{\partial \theta} \pi(\theta | \mathcal{H}_j)$ denotes the gradient, and $[D^2 \pi(\theta | \mathcal{H}_j)] = \left[\frac{\partial^2}{\partial \theta \partial \theta} \pi(\theta | \mathcal{H}_j) \right]$ denotes the Hessian matrix.

Theorem 2 (Asymptotic sampling distribution of a peri-null Bayes factor) *Under the regularity conditions stated in Theorem 1 and for all data-governing parameters θ for which*

$$\dot{v}(\theta) := [D^1 \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right)] \neq 0 \in \mathbb{R}^p, \quad (9)$$

the asymptotic sampling distribution of the logarithm of the peri-null Bayes factor is normal, that is,

$$\sqrt{n} \left(\log \text{BF}_{10}(Y^n) - \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right) - E(\theta, n) \right) \overset{\mathbb{P}_\theta}{\rightsquigarrow} \mathcal{N} \left(0, \dot{v}(\theta)^T I^{-1}(\theta) \dot{v}(\theta) \right), \quad (10)$$

where $\overset{\mathbb{P}_\theta}{\rightsquigarrow}$ denotes convergence in distribution under $\mathbb{P}_\theta$ and where

$$E(\theta, n) = \log \left(\frac{1 + C^1(\theta | \mathcal{H}_1)/n + C^2(\theta | \mathcal{H}_1)/n^2}{1 + C^1(\theta | \mathcal{H}_0)/n + C^2(\theta | \mathcal{H}_0)/n^2} \right), \quad (11)$$

⁴ We thank the second anonymous reviewer for bringing this reference to our attention. When comparing our Theorem 1 to that of Theorem 1 (ii) of Dawid (2011), it is worth noting that the apparent difference in the order of the remainder term vanishes once the MLE $\hat{\theta}$ in Eq. (7) is replaced by $\hat{\theta} = \theta + h/\sqrt{n}$, which holds in probability for large n for regular models.

is a bias term that is asymptotically negligible, and where $C^1(\theta | \mathcal{H}_j)$ and $C^2(\theta | \mathcal{H}_j)$ are given explicitly by Eq. (32) and Eq. (33) based on the notation introduced in Appendix A.

For all θ for which $\dot{v}(\theta) = 0$, but $\ddot{v}(\theta) := [D^2 \log(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)})] \neq 0 \in \mathbb{R}^{p \times p}$, the asymptotic distribution of $\log \text{BF}_{1\tilde{0}}(Y^n)$ has a quadratic form, that is,

$$n \left(\log \text{BF}_{1\tilde{0}}(Y^n) - \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right) - E(\theta, n) \right) \overset{\mathbb{P}_\theta}{\rightsquigarrow} Z^T I^{-1/2}(\theta) \ddot{v}(\theta) I^{-1/2}(\theta) Z, \quad (12)$$

where $Z \sim \mathcal{N}(0, I)$ with $I \in \mathbb{R}^{p \times p}$ the identity matrix. $\diamond$

Proof The proof depends on (another) Taylor series expansion, see Appendix A for full details. Firstly, we recall that $\sqrt{n}(\hat{\theta} - \theta) \overset{\theta}{\rightsquigarrow} \mathcal{N}(0, I^{-1}(\theta))$. To relate this asymptotic distribution to that of $\log \text{BF}_{1\tilde{0}}(Y^n)$, we note that Eq. (8) is, up to a decreasing error in n , a smooth function of the maximum likelihood estimator. The goal is to ensure that the error terms $1 + C^1(\theta | \mathcal{H}_j)/n + C^2(\theta | \mathcal{H}_j)/n^2$ are asymptotically negligible. A Taylor series expansion at the data-governing θ shows that

$$\begin{aligned} \log \text{BF}_{1\tilde{0}}(Y^n) &= \log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right) + E(\theta, n) \\ &+ (\hat{\theta} - \theta)^T (\dot{v}(\theta) + [D^1 E(\theta, n)]) \\ &+ (\hat{\theta} - \theta)^T \frac{(\ddot{v}(\theta) + [D^2 E(\theta, n)])}{2} (\hat{\theta} - \theta) + \mathcal{O}_P(n^{-3/2}). \end{aligned} \quad (13)$$

The asymptotic normality result follows after rearranging Eq. (13), a multiplication of $\sqrt{n}$ on both sides, and an application of Slutsky's lemma.

To conclude that the bias term $E(\theta, n)$ is indeed asymptotically negligible, note that $\log(1 + x/n) \approx x/n$ as $n \rightarrow \infty$ and therefore $D^k E(\theta, n) = \mathcal{O}\left(\frac{1}{n} D^k \{C^1(\theta | \mathcal{H}_1) - C^1(\theta | \mathcal{H}_0)\}\right)$ for all $k \leq 3$. The approximation $\log(1 + x/n) \approx x/n$ requires $C^k(\theta | \mathcal{H}_j)$ for $k = 1, 2$ and $j = \tilde{0}, 1$ to be of similar magnitude, but this is typically not the case when κ_0 is relatively small compared to κ_1 . The bias is, therefore, expected to decay much more slowly.

Similarly, when $\dot{v}(\theta)$ is zero, but $\ddot{v}(\theta)$ not, we have

$$\begin{aligned} n \log \text{BF}_{1\tilde{0}}(Y^n) &= n \left(\log \left(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)} \right) + E(\theta, n) \right) \\ &+ \sqrt{n}(\hat{\theta} - \theta)^T \frac{(\ddot{v}(\theta) + \mathcal{O}(n^{-1}))}{2} \sqrt{n}(\hat{\theta} - \theta) + \mathcal{O}_P(n^{-1/2}). \end{aligned} \quad (14)$$

Since $\sqrt{n}(\hat{\theta} - \theta) \overset{\mathbb{P}_\theta}{\rightsquigarrow} \mathcal{N}(0, I(\theta)^{-1})$, the second order result follows. $\square$

Theorem 2 also shows that under the alternative hypothesis, $\log \text{BF}_{1\tilde{0}}(Y^n)$ is expected to increase towards the limiting value $\log(\frac{\pi(\theta | \mathcal{H}_1)}{\pi(\theta | \mathcal{H}_0)})$ as $n \rightarrow \infty$ whenever $E(\theta, n) < 0$. The bias is expected to be negative, because if the data-governing parameter δ is far from zero, but the peri-null prior is specified such that it is peaked at zero, the Laplace approximations become less accurate. In other words, for fixed n and $\delta \neq 0$, we typically have $C^1(\theta | \mathcal{H}_1) \leq C^1(\theta | \mathcal{H}_0)$